WORKING MANUSCRIPT · FROZEN MODEL-LEVEL EVIDENCE

Asymmetric global assembly and local fidelity in a meta-learned plastic recurrent network

A computational account of coherent and individualized rankings from sparse few-shot evidence

26 August 2026

Central model-level claim

A meta-learned plastic recurrent network transforms sparse, partially encoded signed evidence into coherent yet individualized rankings through two causally distinct routes: selective global admission feeds an interacting fast-weight state P_T for remote and nonlearned relational assembly, whereas broader weak admission feeds a query-addressed local state L_T, exactly represented by edge ledger a_T, for direct experience fidelity.

This manuscript summarizes registered, immutable results. It introduces no new training, replay, resampling, threshold, estimand, or human-neural claim.

Abstract

Humans can infer coherent global rankings after observing only a sparse subset of local pairwise relations, while different individuals exposed to identical evidence may form stable but distinct rankings. This combination of generalization, coherence, and individualization poses a computational challenge: a system must propagate local evidence beyond directly experienced pairs without erasing relation-specific experience. We studied this problem using a meta-learned recurrent network with neuromodulated within-episode plasticity. The network was trained on generic connected ranking graphs while the target eight-item graph and its reflection were held out. Evaluation preserved the human task interface: eight signed support relations, four passive presentations per relation, all-pair testing, and no learning response or test feedback.

The frozen model reproduced six of nine registered behavioral phenomena quantitatively and reproduced the direction of the remaining three, while retaining an excessive symbolic-distance slope, a weak serial-position endpoint contrast, and excess self-inconsistency in one network. Causal interventions identified an asymmetric computation. Selectively admitted evidence accumulated in a recurrent fast-weight state P_T whose removal collapsed remote influence and nonlearned inference. Broader weak evidence entered a persistent local trace L_T whose natural query address selectively rescued directly experienced relations. Removing the local path restored the global-only model exactly. The local computation reduced without approximation to an additive relation ledger a_T and fixed Gram readout, whereas potential-only, scalar-history, item-history, and fixed linear full-state reductions did not close the global learning dynamics. The functional organization transported across prospectively registered changes in support topology, presentation order, evidence density, and item count from six to ten, but quantitative performance and individualization remained condition sensitive.

These results support a model-level division between distributed, history-dependent relational assembly and explicit, query-addressed preservation of direct evidence. They do not establish a human neural implementation, biological memory stores, arbitrary scaling, or a unique compact global algorithm.

Keywords: few-shot learning; relational inference; transitive inference; meta-learning; recurrent plasticity; fast weights; individualized rankings

1. Introduction

Sparse experience often supports conclusions that were never directly observed. Ranking is a particularly transparent example: a learner may see a small set of pairwise relations, yet later answer queries over an entire item set. Classical transitive-inference accounts emphasize how local comparisons support novel choices, and recent theories formalize how relational geometry can generate systematic generalization (Lippl et al., 2024). Knowledge-assembly studies further show that newly observed relations can reorganize an existing relational structure (Nelli et al., 2023). The central computational question is not only whether a system generalizes, but how it constructs a globally coherent field while retaining the local details on which that field is based.

Liu, Wang, and Luo introduced a stringent few-shot ranking paradigm in which participants passively viewed sparse, signed pair evidence and were later tested on all item pairs without feedback (Liu et al., 2026). Identical input did not lead to a single common solution. Participants formed stable and largely self-consistent rankings, but their reconstructed orders differed across individuals and frequently differed from the ground-truth order. Group-level accuracy therefore coexisted with stable participant-specific errors. This phenotype is not captured by asking only whether a model recovers the correct rank; a suitable model must explain coherence, generalization, direct-pair fidelity, and individualization together.

Plastic recurrent networks provide a candidate substrate because they can learn a within-episode learning rule rather than receive a hand-specified relational algorithm. Miconi and Kay showed that meta-learned recurrent plasticity can support transitive inference and fast knowledge reassembly (Miconi & Kay, 2025). Their rewarded adjacent-pair and list-linking tasks establish model ancestry and a useful computational motif, but they do not reproduce the Liu observation interface. The present work therefore treats the Liu task contract as an independent information boundary and tests whether a meta-learned plastic system can acquire a task-faithful solution.

A global relational representation alone creates a tension. A coherent potential can answer unobserved queries, but its compression may redistribute or suppress relation-specific evidence. Conversely, an explicit store of observed edges can preserve direct experience, but it need not infer relations that were never stored. We asked whether these demands are served by distinct computations within one model. We separated five levels of evidence: behavioral competence; direct causal readout; recurrent global rollout; algorithmic sufficiency; and transport beyond the original graph. This separation prevents behavioral resemblance from being mistaken for mechanism and prevents a model intervention from being interpreted as a human neural result.

The resulting account is asymmetric. A high-dimensional recurrent fast-weight state performs history-dependent global assembly, whereas a fixed content-addressable local trace preserves direct evidence. The local trace has an exact low-dimensional relation-ledger description. The terminal global output is nearly additive, but the tested reduced learning states are insufficient. The distinction between simple output geometry and closed learning dynamics is central to the result.

2. Methods

2.1. Study design and separation of learning stages

The neural model was fitted in two non-overlapping stages. First, a plastic recurrent backbone was meta-trained on newly sampled generic ranking episodes. Second, with every backbone tensor frozen, one scalar gain for a content-addressable local trace was learned on the same generic task family. The completed model was then evaluated without parameter updates on the Liu task (Liu et al., 2026). The source-correct Liu support graph and its rank-axis reflection were excluded from both learning stages. Thus, **query labels supervised the generic outer-loop objectives, but no Liu trial or outcome entered backbone or local-gain optimization, and no query label or feedback was ever presented as an episode input**.

A separate response-noise temperature mapped frozen model margins to sampled choices. This was not a neural parameter: it had been selected once on an earlier development checkpoint using approximate human overall accuracy and was subsequently held fixed. We report its selection rule explicitly below rather than treating it as part of meta-training.

We use θ for across-episode backbone parameters. The hidden state h, eligibility trace E, recurrent fast-weight state P, and local trace L are within-episode variables; they are initialized anew for every sampled episode and are never optimizer parameters.

Stage	Task distribution	Optimized quantity	Quantities held fixed
Backbone meta-training	Random eight-item connected sparse graphs	θ	Task generator and held-out graph rule
Local-gain adaptation	New episodes from the same generic family	One raw gain g_L	Final backbone and local address/update
Earlier choice calibration	Approximate human overall accuracy only	One grid value T_{choice}	Neural model and all secondary metrics
Frozen evaluation	Liu eight-relation protocol	None	θ, g_L, T_{choice}, task interface, and analysis rules

Table 1: **Separation of learning, choice calibration, and evaluation.** Neural checkpoints and local gains were never selected on Liu behavior. The earlier descriptive choice-temperature selection was not repeated on the fresh networks.

2.2. Generic meta-training task

An episode defined a latent total order over $N = 8$ items. We write $\pi = (\pi_1, \ldots, \pi_N)$ from highest to lowest and $r(i)$ for the rank position of item i. Each item received a newly sampled cue $c_i \in \{-1, +1\}^C$ with $C = 15$. To prevent nearly duplicate cues, any pair of accepted codes agreed on at most 66% of coordinates. The item order, cue set, support graph, presentation order, orientation, and subject encoding state were sampled independently for each episode.

At the start of each meta-batch, the number of support edges K was sampled uniformly from $\{7, 8, 9, 10\}$. Each of the 32 episodes in that batch then received an independent connected graph with K edges over rank positions. A graph was accepted only if it contained at least two rank gaps and did not match the Liu graph or its reflection. Each edge was presented once in each of four support blocks, giving $S = 4K$ passive support trials. Edge order was randomized within block, and left/right orientation was sampled anew on every presentation.

For an oriented support pair with rank gap $\delta_{ij} = |r(i) - r(j)|$, the task-visible signed magnitude was

$$m_j = a_j \frac{\delta_{ij}}{N-1}, \quad a_j = \begin{cases} +1 \\ \text{higher item on the left} \\ -1 \\ \text{higher item on the right} \end{cases}.$$

The episode also contained a stable relation-encoding bottleneck. Its latent variables were sampled once per episode as $b \sim \mathcal{N}(1.0, 0.5^2)$, $u_i \sim \mathcal{N}(0, 0.35^2)$, and $\beta \sim \mathcal{N}(0, 0.25^2)$. With $p_{\min} = 0.1$ and logistic function $\sigma(v) = \frac{1}{1+\exp(-v)}$, relation reliability was

$$p_{ij} = p_{\min} + (1 - p_{\min})\sigma\left(b + u_i + u_j + \beta\left[\frac{\delta_{ij}}{N-1} - \frac{1}{2}\right]\right).$$

One retention variable $z_{ij} \sim \text{Bernoulli}(p_{ij})$ was drawn for each support relation and reused across its four presentations. The recurrent path therefore received signed evidence $s_j^G = m_j z_{ij}$: a retained relation entered at its full task-visible magnitude, whereas an omitted relation entered as zero.

After support, all $Q = \binom{8}{2} = 28$ unordered pairs were queried once in a random order and orientation. Target $y_q = 1$ meant that the left item was higher; $y_q = 0$ meant that the right item was higher. The target was supplied only to the outer cross-entropy loss. Query inputs contained neither y_q, signed evidence, rank position, symbolic distance, nor feedback.

2.3. Trial input, timing, and state persistence

Every support trial comprised four recurrent steps $r = 0, 1, 2, 3$. With I denoting an indicator, the 37-dimensional input at support trial j was the concatenation

$$x_{jr} = \left(I_{r=0} c_l, I_{r=0} c_r, I_{r=1}, 1, \tau_j, 0, I_{r=0} s_j^G, 0, 0\right) \in \mathbb{R}^{37}.$$

The first 30 entries are the ordered pair cues; the next entry is a response pulse; the following four are bias, normalized episode time, reward, and signed-evidence channels; the final two are inherited action channels. Reward and action channels were always zero because support was passive and no previous action was fed back. Although the recurrent cell emitted logits on every step, all support logits were discarded: they generated neither a behavioral response nor a loss. The time value was constant within a trial and increased linearly from zero on the first support trial to $\frac{2}{3}$ on the last: $\tau_j = \left[\frac{j-1}{S-1}\right]\left(\frac{2}{3}\right)$. It was held at $\frac{2}{3}$ during query.

A query used the same cue and response-pulse sequence but set the evidence channel to zero. The choice logits at $r = 1$ defined the response, so only steps $r = 0, 1$ were required for the training loss. At the start of every support or query trial, h and E were reset to zero. The fast-weight state P began at zero, persisted across all support trials, and was then frozen at its terminal value P_T for every independently initialized query. Consequently, neither query order nor a previous query response could alter a later query.

2.4. Plastic recurrent backbone

The backbone was a one-layer recurrent network with hidden size $H = 200$ and neuromodulated within-episode recurrent plasticity (Miconi & Kay, 2025). At a recurrent step, the hidden state and the two action logits were

$$h_r = \tanh(W_{\text{in}} x_r + b_h + [W + \alpha \odot P_{r-1}] h_{r-1}), \quad o_r = W_o h_r + b_o.$$

Here $W \in \mathbb{R}^{H \times H}$ is the structural recurrent matrix, $\alpha \in \mathbb{R}^{H \times H}$ is a learned elementwise plastic sensitivity, and P_{r-1} is the episode-specific fast-weight matrix. A two-unit modulatory head generated

$$q_r = \tanh(W_{\text{DA}} h_r + b_{\text{DA}}), \quad d_r = D_{\text{mult}}(q_{r,1} - q_{r,2}).$$

The fast-weight and eligibility updates were then applied in the following order:

$$\tilde{P}_r = P_{r-1} + d_r E_{r-1}, \quad P_r = \Pi_{\mathcal{B}_{50}}(\tilde{P}_r),$$

$$E_r = (1 - \eta) E_{r-1} + \eta \tanh(h_r h_{r-1}^T).$$

$\mathcal{B}_c$ denotes the elementwise box $\{A \in \mathbb{R}^{H \times H} : |A_{uv}| \le c \ \text{ for every } \ u, v\}$, and $\Pi_{\mathcal{B}_c}$ is its elementwise projection. Thus the second equation is a mathematically defined bounded update, not an undefined software `clip` operation. The implemented order is $h_r \to q_r, d_r \to P_r \to E_r$: the current modulation multiplies the **previous** eligibility trace, and the current eligibility enters only a later plastic update.

The meta-learned backbone parameters were $W_{\text{in}}, b_h, W, \alpha, W_o, b_o$, the modulatory head, D_{mult}, and η. The checkpoint retained a scalar value head inherited from the ancestral reinforcement-learning architecture, but its output entered neither the supervised objective nor any reported readout and therefore received no task gradient in the present training protocol.

2.5. Outer-loop optimization of the backbone

For a meta-batch of $B = 32$ episodes, let o_{bq} be the response logits for query q in episode b, and let P_T^b be its terminal support state. The backbone objective was

$$\mathcal{L}_{\text{backbone}} = \frac{1}{BQ}\sum_{b=1}^{B}\sum_{q=1}^{Q}\text{CE}(o_{bq}, y_{bq}) + \lambda_P \frac{1}{BH^2}\sum_{b=1}^{B}\left\|P_T^b\right\|_F^2, \quad \lambda_P = 10^{-4}.$$

Gradients were propagated through the support rollouts, plastic updates, and query readouts into θ; the sampled task variables and within-episode states were not free parameters. Each fresh backbone was trained for exactly 1,000 outer updates with Adam (learning rate 10^{-4}, moment coefficients 0.9 and 0.999, numerical epsilon 10^{-8}, and no weight decay), global gradient-norm clipping at 2.0, and no early stopping. Networks 2104 and 2105 used independent declared random seeds. Only the step-1,000 checkpoint was retained for gain adaptation and evaluation; intermediate behavior did not select a checkpoint.

2.6. Content-addressable local trace and gain adaptation

The local path stores directly presented relations without changing the backbone. For ordered cues (c_l, c_r), its antisymmetric conjunctive address was

$$k(c_l, c_r) = \frac{\text{vec}(c_l c_r^T - c_r c_l^T)}{\max\left(\left\|\text{vec}(c_l c_r^T - c_r c_l^T)\right\|_2, 10^{-8}\right)} \in \mathbb{R}^{C^2}.$$

The address reverses sign when cue order reverses. Starting from $L_0 = 0 \in \mathbb{R}^{225}$, the trace received exactly one write at the first step of each support trial,

$$L_j = L_{j-1} + s_j^L k(c_l, c_r).$$

For query q, the raw local margin was $\ell_q = L_T^T k(c_l, c_r)$. If $o_{q,0}$ and $o_{q,1}$ are the recurrent logits, the combined logits were

$$\tilde{o}_{q,0} = o_{q,0} - \frac{1}{2}\lambda_L \ell_q, \quad \tilde{o}_{q,1} = o_{q,1} + \frac{1}{2}\lambda_L \ell_q,$$

so the left-versus-right margin became $(o_{q,1} - o_{q,0}) + \lambda_L \ell_q$. Positivity was enforced by $\lambda_L = \text{softplus}(g_L)$.

The local gain was learned **after** backbone training. During this generic-only adaptation stage, local and recurrent paths shared the same admitted value, $s_j^L = s_j^G = m_j z_{ij}$. All backbone tensors and the local address/update were frozen; only g_L was optimized. The loss was mean cross-entropy over all 28 queries of the combined global-plus-local readout, using batches of 32 new generic episodes. Adaptation ran for exactly 500 Adam updates (initial $\lambda_L = 0.1$, learning rate 0.01, gradient-norm limit 2.0), with no auxiliary target, Liu trial, or checkpoint selection. The final gains were 0.1521 and 0.1632 for networks 2104 and 2105, respectively.

2.7. Frozen Liu evaluation and differential evidence admission

The held-out evaluation instantiated the eight labeled roles with true order $H > G > F > E > D > C > B > A$. The source-correct support relations were F>A, C>B, E>B, G>C, F>D, G>D, H>E, and H>A. Each appeared once in each of four blocks, yielding 32 support trials. Relation order and orientation were randomized within block, and the visible magnitude remained the true rank gap divided by $N - 1$. Random absolute display height was excluded as nuisance, but relative magnitude was retained as participant-available evidence. Support remained passive: neither humans nor models made a learning response or received feedback.

Each network was evaluated on 77 virtual participants. They shared one deterministically generated set of eight sufficiently distinct bipolar cues, but the item-to-cue assignment was independently permuted across participants. Each participant also received one episode-level encoding state and one stable z_{ij} per support relation under the reliability model defined above. The global path retained its training-time admission rule $s_j^G = m_j z_{ij}$. The confirmed differential local rule was

$$s_j^L = m_j\left[z_{ij} + (1 - z_{ij})p_{ij}\right].$$

Therefore, a retained relation ($z_{ij} = 1$) wrote the identical value to both paths, whereas an omitted relation ($z_{ij} = 0$) wrote zero globally and the reliability-weighted value $m_j p_{ij}$ locally. This rule introduced no new trainable or Liu-fitted parameter; the gain remained the generic-only value fixed above.

After support, P_T and L_T were fixed. All 28 unordered pairs were queried in each of ten blocks, for 280 choices per participant, with randomized orientation and no feedback. Let $M_q = \tilde{o}_{q,1} - \tilde{o}_{q,0}$. Exact left-choice probability and sampled behavior used the fixed temperature $T_{\text{choice}} = 0.25$:

$$\Pr(\text{choose left} \mid q) = \sigma\left(\frac{M_q}{T_{\text{choice}}}\right).$$

The temperature was a descriptive calibration inherited from an earlier development network. It was selected from the finite grid $\{1.0, 0.75, 0.5, 0.25\}$ by minimizing absolute error to an approximate released human overall accuracy of 0.87. The grid had been frozen only after inspecting the 1.0 and 0.5 settings; consequently, this was not a confirmation-stage fit. Learned accuracy, nonlearned accuracy, distance slope, ranking classes, stable errors, pair classes, and inter-subject similarity did not enter selection. The selected value was never refitted for networks 2104 or 2105.

All checkpoints, gains, admission equations, task schedules, and analysis rules were frozen before either fresh network was exposed to this evaluation.

2.8. Behavioral estimands

Learned pairs were the eight support relations; the remaining 20 unordered pairs were nonlearned. Accuracy was averaged within participant and then summarized within dataset. Symbolic-distance slope was computed on the frozen distance profile. Serial-position accuracy averaged the seven incident pairs for each true rank position; endpoint contrast was

$$C_{\text{end}} = \frac{a_1 + a_8}{2} - \frac{1}{6}\sum_{r=2}^{7} a_r.$$

Pair-distribution classes, stable-error prevalence, self-consistency, and analysis eligibility followed the released-paper-aligned frozen specification. The 80% stable-error criterion required the same incorrect choice on at least 80% of repeated queries for a pair.

For Hodge reconstruction, the complete antisymmetric pair field g_{ij} was projected onto the zero-sum gradient subspace:

$$\hat{s} = \operatorname{argmin}_{\sum_i s_i = 0} \sum_{i<j} \left[g_{ij} - (s_i - s_j)\right]^2.$$

Sorting $\hat{s}$ produced the reconstructed subjective ranking. Pairwise Kendall correlation among eligible non-correct orders measured inter-subject similarity. A nearly unit Hodge-gradient fraction indicates a coherent additive field; it does not measure the dimensionality of hidden or fast-weight state.

2.9. Causal estimands and controls

Fast-weight necessity used reset, write-off, plastic-sensitivity-off, and shuffle controls. Relation leave-one-out interventions removed one registered support write and measured direct, disjoint, and third-party consequences. Factor swaps separated eligibility direction, modulatory magnitude, and plastic-sensitivity placement. History-matched comparisons tested whether accumulated state changed later expression.

Local direct fidelity used exact correct-choice probability and the correct-signed direct component of the relation-LOO Hodge residual. Evidence specificity compared natural routing with a signed-scalar multiset derangement within participant and support block. Query specificity retained the natural trace but applied a canonical within-participant derangement of query addresses. Both controls preserved scalar values, write counts, timing, gain, and the recurrent path.

Global-off replaced P_T with zeros during query while retaining L_T. Local-off set the applied local gain to zero without changing P_T, the query rollout, or the stored local state, and therefore restored the backbone readout exactly. Together, the P-off and local-off conditions provided the causal double dissociation.

2.10. Replication, uncertainty, and network scope

The global formal mechanism program used ten unfiltered networks 2001–2010. Local-trace replication used independent development networks 2102 and 2103. Differential-access confirmation then used fresh networks 2104 and 2105 after checkpoints and gains were jointly locked. The behavioral map and paper-aligned figures use only the fresh confirmation outputs.

Participant bootstrap used 10,000 resamples within each network and condition where registered. Participants were never pooled across networks, and networks were not bootstrapped as a population. Cross-network arithmetic means, when reported in source documents, are descriptive. A link passed only under its frozen within-network interval and conjunction rule.

2.11. Transport designs

Topology transport held the eight-item rank-distance multiset fixed and tested three deterministically selected non-isomorphic support graphs. Order transport permuted the identical 32-event evidence multiset into blockwise random, relation-clustered, and exact reverse schedules. Density transport used connected eight-item graphs with seven through ten observed relations. Item-count transport used prospectively selected cycle graphs at $N = 6, 8, 10$, with four presentations per relation and all-pair query. The $N = 6$ and $N = 10$ conditions were out of distribution for backbones trained only at $N = 8$.

Every transport test kept the checkpoint, local gain, evidence-admission equation, cue construction, recurrent update, local address, activation, output readout, temperature, participant rules, and decision thresholds fixed. Transport was sequential and one-factor; no factorial combination was evaluated.

3. Results

3.1. A task-faithful model reaches behavioral competence without reproducing every human statistic

The frozen task contained eight items and eight signed support relations. Each relation was passively presented four times, yielding 32 support trials. Relative displayed magnitude was retained as task information; randomized absolute height was treated as nuisance. Participants and virtual participants made no response during support and received no support or test feedback. The test phase queried all 28 unordered pairs over ten blocks. True rank, query labels, symbolic distance, and feedback never entered the episode input.

The recurrent backbone was meta-trained on generic eight-item connected sparse graphs containing seven to ten edges. The source-correct Liu graph and its rank-axis reflection were prospectively excluded from training. Each virtual participant had one stable evidence-encoding state for the episode. This state changed which observed relations entered the model; it did not contain the true rank or a query target.

The final frozen behavioral map compared the released human benchmark with two fresh networks, 2104 and 2105, without pooling virtual participants or networks. Learned-pair accuracy was 0.914 in humans and 0.925/0.931 in the two networks. Nonlearned accuracy was 0.828 in humans and 0.804/0.805 in the networks. Thus the model retained above-chance direct evidence and substantial inference over unobserved pairs at human-range cohort means (Figure 1).

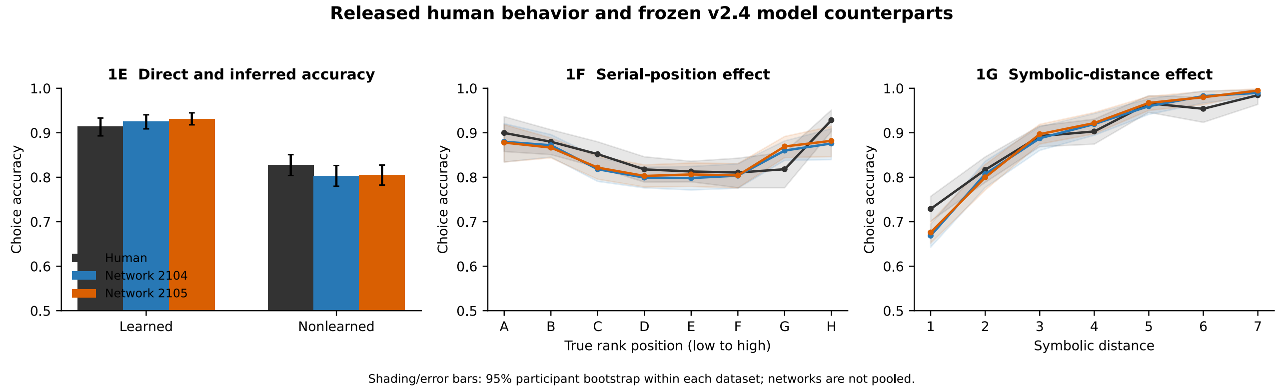

Figure 1: **Released human group behavior and frozen model counterparts.** Learned and nonlearned accuracy, serial-position profiles, and symbolic-distance profiles are shown for the released human data and independent networks 2104 and 2105. Shading and error bars are 95% participant-bootstrap intervals computed separately within each dataset. The model matches cohort accuracy while retaining a weaker endpoint contrast and steeper symbolic-distance slope.

The comparison was frozen as a nine-phenomenon map. Six phenomena were quantitatively reproduced and three were qualitatively reproduced but quantitatively mismatched; none was absent. The retained mismatches were not treated as tuning objectives.

Registered phenomenon	Human	Network 2104	Network 2105
Learned accuracy	0.914	0.925	0.931
Nonlearned accuracy	0.828	0.804	0.805
Symbolic-distance slope	0.0398	0.0495 · steep	0.0495 · steep
Endpoint contrast	0.0822	0.0527 · weak	0.0514 · weak
Pair Beta classes	15 bimodal / 13 high	18 / 10	15 / 13
At least one 80% stable error	0.913	0.944	0.945
Incorrect ranking classes	64 consistent / 5 inconsistent	59 / 12 · excess	65 / 8
Correct Hodge ranking	8 / 77	6 / 77	4 / 77
Mean inter-subject rank tau	0.554	0.535	0.536

Table 2: **Frozen nine-phenomenon behavioral map.** Numeric values are displayed point estimates. Status decisions used the registered human intervals and were made separately for each network. The three marked discrepancies remain active model constraints.

The two shape discrepancies point in opposite amplitude directions: symbolic-distance dependence is too strong, whereas the endpoint advantage is too weak. A single output temperature or global gain cannot repair both without changing the estimand. In addition, network 2104 produced 12 self-inconsistent virtual participants among 77, above the frozen human upper bound, whereas network 2105 did not. The complete behavioral statement is therefore competence with explicit quantitative limits, not a perfect fit.

3.2. Pair-level behavior is stable, coherent, and individualized

Group means do not reveal the defining individual structure. The released human data contained pair types that were consistently easy and pair types with strongly polarized participant responses. Both fresh networks reproduced this mixture, including pairwise bimodality and high-accuracy classes (Figure 2). The model was not calibrated pair by pair to the human matrix; the panels place frozen model outputs on the released paper's estimands.

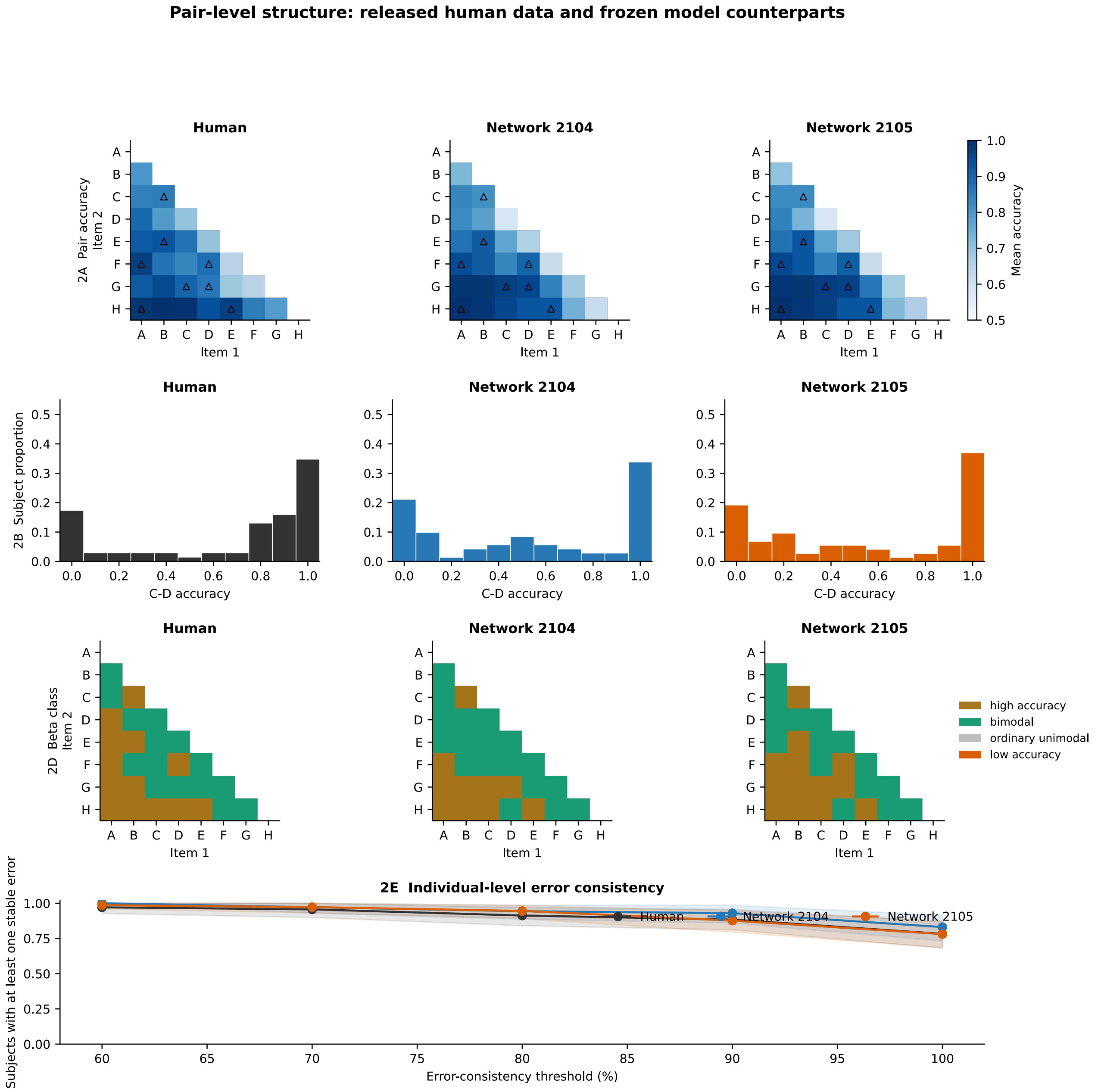

Figure 2: **Pair-level structure in released human data and frozen model outputs.** The panels compare mean pair accuracy, a representative bimodal pair, pair-distribution classes, and the prevalence of stable errors across consistency thresholds. Triangles mark the eight directly observed support relations. Human participants and virtual participants are never pooled across datasets.

Errors were not transient response noise. At the registered 80% threshold, 91.3% of human participants and 94.4%/94.5% of virtual participants had at least one pair on which the same error recurred. Subject-by-pair error maps show sparse but persistent fingerprints in all three datasets (Figure 3). Eligibility counts differ because the registered panel excludes correct rankers and applies the frozen analysis filters.

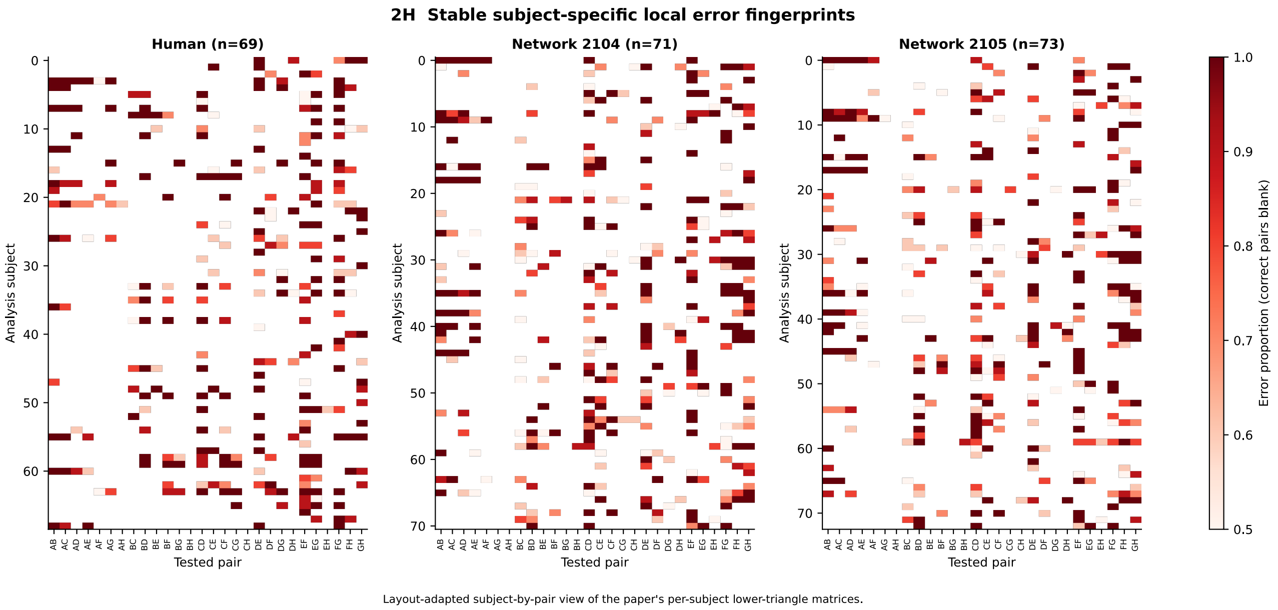

Figure 3: **Stable subject-specific error fingerprints.** Rows are eligible analysis participants and columns are tested pairs; color denotes the proportion of incorrect responses for cells at or above 0.5. The panel is a layout-adapted rendering of the released human analysis and matched frozen model fields, not a cross-subject alignment.

Hodge least-squares reconstruction converted each complete antisymmetric pair field into a zero-sum item potential and a resulting order. Only 8 of 77 human participants and 6/4 of 77 virtual participants in networks 2104/2105 recovered the ground-truth order. Most remaining orders were nevertheless self-consistent. Mean inter-subject Kendall correlation was 0.554 in humans and 0.535/0.536 in the two networks. The model therefore generated coherent global structures without collapsing all participants to one ranking (Figure 4).

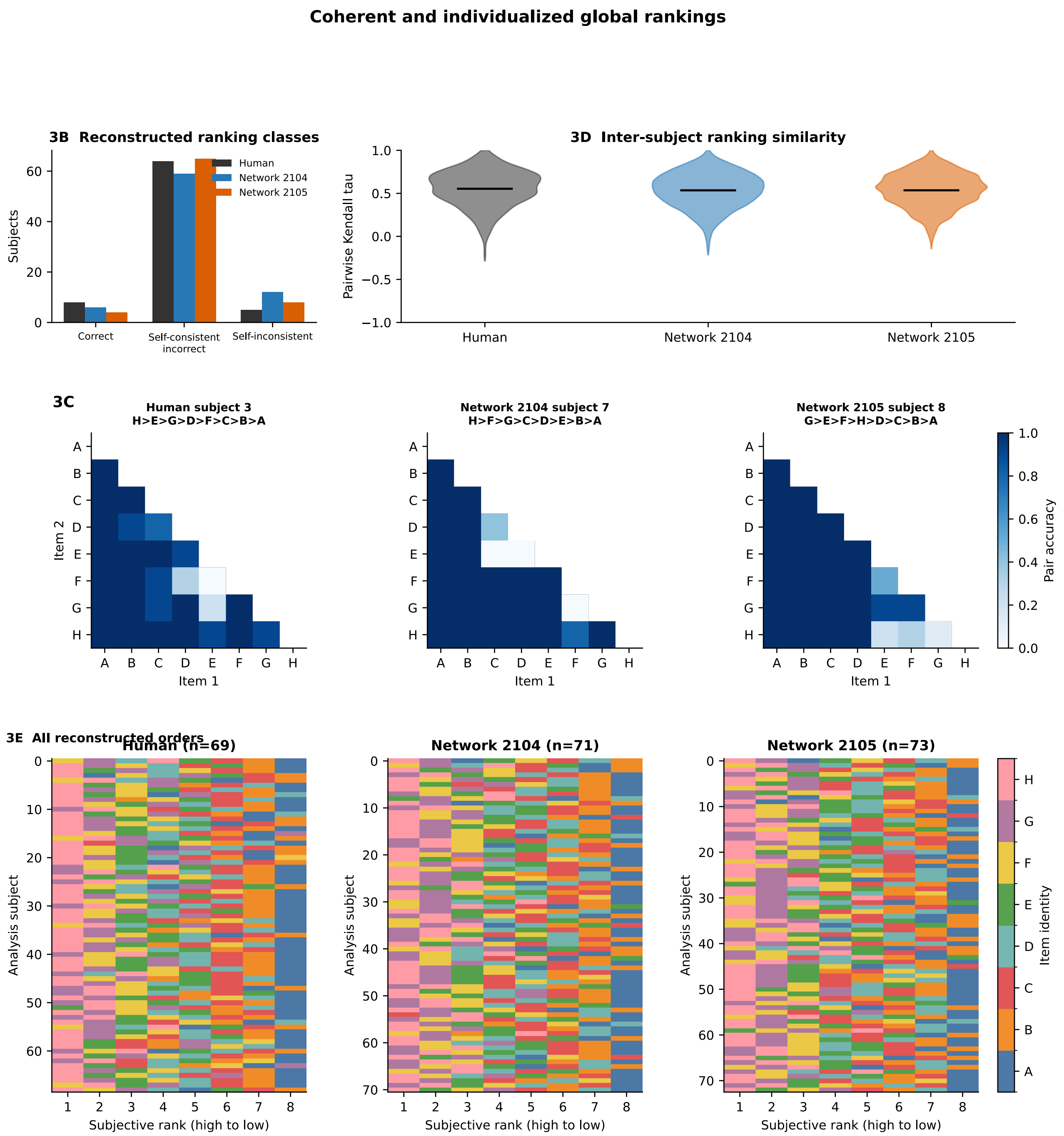

Figure 4: **Coherent and individualized reconstructed global rankings.** Ranking-class counts, inter-subject similarity, representative pair matrices, and all eligible reconstructed orders are shown for humans and the two fresh networks. Representative subjects illustrate structure rather than one-to-one matching across datasets.

These results establish the phenotype to be explained. They do not identify the internal computation. In particular, a model could generate coherent pair fields through a direct score writer, a lookup table, or a post hoc projection while bypassing recurrent plasticity. We therefore next required causal evidence for how observed support changed remote and direct queries.

3.3. A recurrent fast-weight state constructs remote relational consequences

The global backbone is a recurrent network with a within-episode plastic matrix P_t. Its hidden state uses the effective recurrent weight $W + \alpha \odot P_t$. A learned modulatory output scales the current eligibility trace before it is added to P_t. Hidden state and eligibility are reset between trials, whereas P_t persists through support and is frozen during query. Consequently, episode-specific relational information can be retained without changing the structural parameters.

Formal confirmation across ten unfiltered networks established six of seven registered global mechanism links. Resetting P_t, disabling its writes, or setting the plastic sensitivity α to zero removed episode-specific competence. Relation-matched write ablations affected disjoint and third-party pairs, establishing remote causal reach rather than replay of the ablated edge. Eligibility-factor swaps transferred relation-specific direction in nine of ten networks; the competent seed-2009 exception was retained. Modulatory gain, natural sensitivity placement, history effects, and expected-rank-over-MAP projection confirmed. The universal modulator-direction criterion did not confirm because of seed heterogeneity.

The terminal global margin field was almost entirely additive:

$$m_{ij}^{G(P_T)} \approx s_i - s_j.$$

This geometry provides a coherent ranking potential, but it does not mean that one support event performs an independently correct Bayesian update. Isolated writes had robust remote magnitude while correctness propagation was negative or unresolved, and the full formal direction-preservation chain was heterogeneous. The supported description is a state-dependent iterative relaxation in which evidence interacts with accumulated fast weights.

Global causal boundary

P_T is necessary for episode-specific remote and nonlearned relational assembly. The evidence does not identify a biological dopamine signal, assign meaning to individual fast-weight entries, or establish exact sequential Bayesian updating.

3.4. Differential evidence admission separates global assembly from direct fidelity

A global-only system was insufficient because an almost additive field can remain coherent while losing directly experienced relation-specific value. The confirmed model therefore uses two evidence-admission rules. For support magnitude m_t, stable global admission z_{sr}, and relation reliability p_{sr},

$$s_t^G = m_t z_{sr}, \quad s_t^L = m_t[z_{sr} + (1 - z_{sr})p_{sr}].$$

The first rule supplies selective effective evidence to the recurrent global state. The second preserves the same retained evidence and adds a weaker, zero-parameter contribution for evidence omitted from the global branch. The reliability state is fixed for the participant and episode. It is a model hypothesis about encoding, not stimulus information or an inferred human latent variable.

The local branch binds the two normal item cues through a fixed normalized antisymmetric conjunctive key. It accumulates one signed value per support event and is read only at the current query address. Its gain was adapted on generic ranking episodes while the recurrent backbone remained frozen; Liu data were not used to fit the gain.

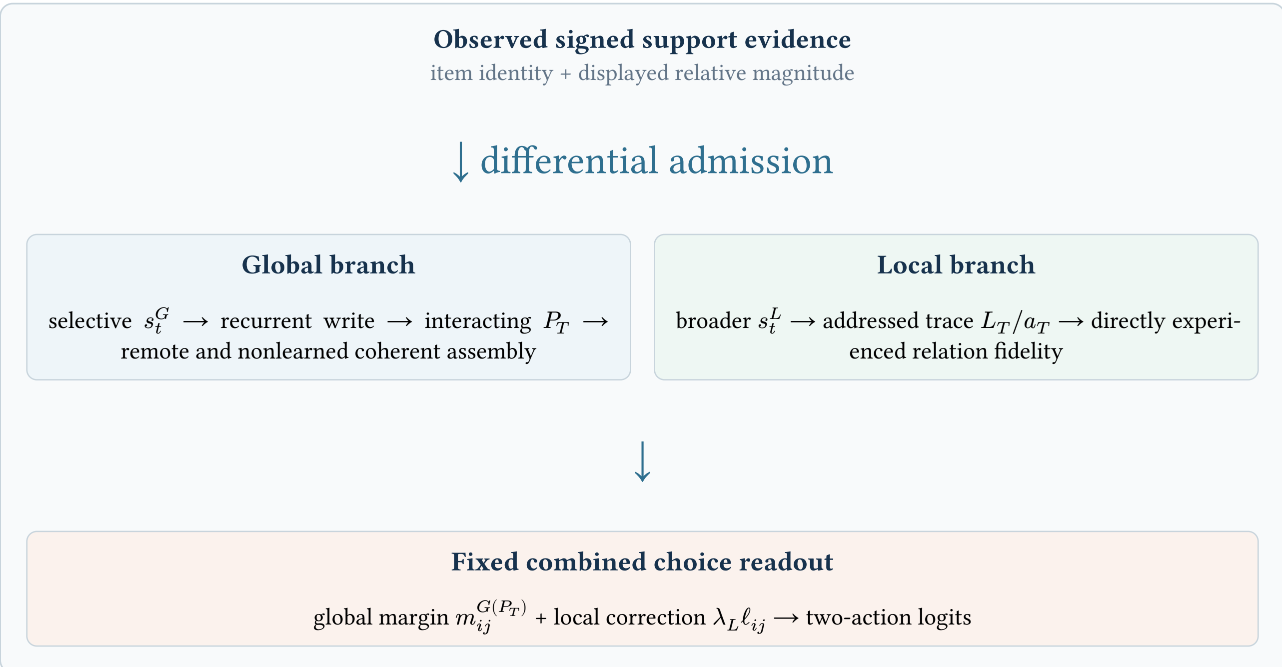

Figure 5: **Confirmed asymmetric computation.** Differential admission routes observed evidence to a history-dependent global fast-weight computation and an additive query-addressed local trace. Both branches contribute to one fixed choice readout.

The causal contrast was replicated first on independent backbones 2102 and 2103 and then confirmed unchanged on fresh backbones 2104 and 2105. In the fresh confirmation, broader local admission improved exact probability on stable-omitted direct queries by 0.061 and 0.052 and improved the registered relation-LOO direct-correctness contrast by 0.194 and 0.208. Natural evidence-to-relation routing and natural trace-to-query addressing each beat their frozen derangements.

Removing P_T preserved a local learned advantage but left nonlearned sampled accuracy near chance; the upper confidence bounds were below 0.47. The registered remote-collapse contrasts were −0.068 and −0.069. Conversely, removing the local contribution restored the global-only v1 logits exactly and preserved every global qualification and mechanism gate. This double dissociation identifies two computations rather than one branch serving as a nonspecific gain.

The local rescue was not free. Broader omitted writes caused small replicated cross-talk on retained exact probability: −0.0011 and −0.0014 in the two fresh networks. Both effects passed the frozen noninferiority margin of −0.005, but retained behavior is not literally unchanged. The local branch also did not repair the excessive nonlearned symbolic-distance slope.

Fresh-backbone contrast	2104	2105	Interpretation
Omitted exact rescue	+0.061	+0.052	direct benefit
Omitted relation-LOO rescue	+0.194	+0.208	direct causal benefit
Natural minus query shuffle	+0.204	+0.219	address specificity
P-off remote contrast	−0.068	−0.069	remote collapse
Retained exact trade-off	−0.0011	−0.0014	small cross-talk

Table 3: **Key contrasts in the fresh-backbone v2.4 confirmation.** Decisions used participant-bootstrap intervals within each network. Values are not pooled network-level estimates.

3.5. The local computation reduces exactly, but tested global learning states do not

The implemented local key for left and right item cues c_l and c_r is

$$k(c_l, c_r) = \frac{\operatorname{vec}\left(c_l c_r^T - c_r c_l^T\right)}{\max\left(\left\|\operatorname{vec}\left(c_l c_r^T - c_r c_l^T\right)\right\|, 10^{-8}\right)}.$$

The tensor/vector implementation accumulates

$$L_{t+1} = L_t + s_t^L k_{r_t}, \quad \ell_q = k_q^T L_T.$$

Because the keys are fixed, the same state can be represented by a ledger a_t over support relations:

$$a_{t+1} = a_t + s_t^L e_{r_t}, \quad L_T = \sum_r a_{T,r} k_r, \quad \ell_q = \left(K a_T\right)_q,$$

where $K_{qr} = k_q^T k_r$. This is an exact change of coordinates, not a second memory store or a fitted surrogate. At the registered eight-item report point, the maximum tensor reconstruction discrepancy was 5.55×10^{-16} and the maximum all-query read discrepancy was 7.99×10^{-15}. Conservative maxima across the item-count transport study remained below 9×10^{-15}.

The global branch showed the opposite relationship between representation and dynamics. Its terminal output was nearly a rank potential, yet a potential-only transition failed prospective closure. Adding scalar history recovered remote update amount but misallocated the field. An item-history state improved allocation but lost the correct remote amount. Finally, a fixed linear readout of the full residual fast-weight state failed held-out prediction and worsened remote allocation. These negatives do not prove that no compact nonlinear global state exists. They establish that no tested reduced state is currently sufficient.

Property	Global branch	Local branch
Implemented state	interacting fast-weight matrix P_t	addressed trace L_t
Update	history-dependent recurrent write	additive signed relation write
Causal role	remote and nonlearned assembly	direct/query-matched fidelity
Terminal output	approximately potential-like	relation-specific Gram read
Reduction result	tested compact states insufficient	exact edge ledger a_t
Order sensitivity	quantitatively order sensitive	exactly invariant to registered permutations

Table 4: **Algorithmic asymmetry.** Simple terminal global geometry does not imply closed low-dimensional learning dynamics; the local implementation admits an exact coordinate reduction.

Claim boundary

The result supports "simple output geometry does not imply simple learning dynamics." It does not support "the global algorithm is intrinsically high-dimensional" or "no compact global state exists."

3.6. The functional organization transports, but performance is not invariant

Transport was tested prospectively along four one-factor axes. The local and global equations, checkpoints, gains, evidence-admission rule, readout, participant construction, and bootstrap rules were frozen. Each decision was made within graph, schedule, density, or size and within backbone; participants and networks were never pooled.

Three matched alternative eight-item topologies passed all eight functional links in all three development backbones (9 of 9 cells; 72 of 72 link decisions). Blockwise-random, relation-clustered, and exactly reversed presentation schedules likewise passed all links in all three backbones. The local ledger was exactly order invariant, whereas the global P_T field changed quantitatively, especially under relation clustering.

Evidence-density transport retained every causal and exact-compression link from seven through ten observed edges. The full parent conjunction passed 23 of 24 cells: one $E = 10$ seed missed the frozen individualized-stability interval. A subsequent registered localization showed that whole subjective orders converged and stable-error counts declined with density, but binary loss of "at least one stable error" did not replicate. The parent outcome therefore remains unresolved rather than being relabeled as a pass.

Item-count transport tested $N = 6, 8, 10$ without retraining. All eight primary links passed in all nine size-by-backbone cells. The local/global organization therefore survived both strict out-of-distribution sizes. Performance was not scale invariant: nonlearned exact accuracy declined from 0.825–0.833 at $N = 6$ to 0.749–0.759 at $N = 10$, Hodge-order correlation declined to 0.585–0.608, remote effect decreased, and normalized distance slope increased.

Axis	Frozen outcome	Exact boundary
Support topology	mechanism transported	three matched $N = 8$ alternatives, not arbitrary graphs
Presentation order	mechanism transported	three schedules; P_T changes quantitatively
Evidence sparsity	dependent or unresolved	one $E = 10$ individualized-stability interval misses
Sparsity localization	order convergence without replicated binary loss	convergence does not rewrite the parent outcome
Item count	mechanism transported	$N = 6, 8, 10$ cycle family; performance degrades with N

Table 5: **Registered one-factor transport outcomes.** Transport concerns the functional division between global and local computation, not invariance of every behavioral metric.

No topology-by-order-by-density-by-size factorial experiment was performed. The transport results show that the functional organization is not a lookup table for one graph or presentation schedule. They do not guarantee arbitrary graphs, arbitrary timing, arbitrary evidence density, or indefinite scaling.

4. Discussion

The central result is a division of computational labor within one meta-learned relational system. Selectively admitted evidence enters an interacting fast-weight state that assembles remote, nonlearned, and nearly potential-like relational structure. Broader weak evidence enters an additive local trace that is read only by the matching query. The first branch provides coherence and generalization; the second preserves direct experience. Causal interventions, routing derangements, exact reduction, and one-factor transport converge on this account.

This division resolves a tension in sparse relational learning. A single global potential is efficient for inference but can discard or redistribute edge-specific evidence. A pure edge store retains observations but does not construct unseen relations. The model combines these roles without requiring the local path to become another transitive learner. Under P-off, direct evidence remains accessible while remote inference collapses. Under local-off, the global system remains intact while the direct-fidelity benefit vanishes.

The algorithmic asymmetry is equally important. The local tensor implementation looks high-dimensional, yet its computation is exactly an edge ledger followed by a fixed Gram operator. By contrast, the global terminal field is low-dimensional in form, but the tested potential, scalar-history, item-history, and full-state linear coordinates do not predict its learning dynamics prospectively. Representation dimensionality and algorithmic state sufficiency are therefore distinct questions.

The model also provides a qualified account of individualization. Stable participant-specific evidence admission produces coherent but different global structures and stable local errors. This is a sufficient model hypothesis, not an estimate of human encoding reliability. The same behavioral phenotype could in principle arise from other latent sources. Human neural implementation, biological storage, and the relationship between model states and frontoparietal MEG representations remain outside the present evidence.

Negative results constrain the theory rather than being averaged away. The symbolic-distance slope is too steep, the serial-position endpoint contrast is too weak, and one network overproduces self-inconsistent rankings. Because the first two mismatches have opposite amplitude directions, they do not motivate a single gain or temperature repair. Density does not reduce dependence on the global path as originally predicted. Larger item sets preserve the local/global division while degrading global quantitative quality. These facts limit the current claim and identify new questions only if a separate prospective program is opened.

Several further limits are explicit. First, fresh-backbone confirmation is not a network-population prevalence estimate. Second, one-factor transport is not a factorial generalization result. Third, the conjunctive address and reliability equation are sufficient implementations, not unique ones. Fourth, failed global reductions do not establish irreducibility in principle. Fifth, behavioral competence licenses model-mechanism analysis but does not turn the model into a description of the human brain.

The frozen result suggests three distinct future directions, none of which is part of the present claim. A new model program could study why global policy quality is cardinality sensitive. A task-generalization program could test list linking under a separate registered contract. A human program could test the predicted separation between direct-evidence fidelity and global reassembly with new behavioral or neural data. The present work stops before those extensions so that the reported evidence object remains stable.

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